Risk Measurement Exercises: Value-at-Risk (VaR) and Expected Shortfall (ES)

Context

This document contains exercises designed to build intuition and technical skill for **Value-at-Risk (VaR)** and **Expected Shortfall (ES)**. No prior knowledge of ERM is required.

Exercise 1 — Interpreting Value-at-Risk (Conceptual)

A trading desk reports a **1-day VaR at 99% confidence** of **€10 million**.

1. Explain in plain language what this number means.
2. Out of 100 trading days, on how many days should losses exceed €10 million?
3. Does this measure tell you how large losses can be beyond €10 million?
4. Can the loss exceed €10 million? Explain.

Exercise 2 — Historical VaR

You observe the following daily portfolio returns (in %) over 10 days:

-3.2 -2.5 -1.8 -1.2 -0.9 -0.4 0.1 0.4 0.7 1.0

1. Sort the returns from worst to best.
2. Compute the **95% historical VaR** for **losses** if portfolio value is €50 million.
3. State clearly which observation(s) determine the 95% historical VaR in this small sample.

Exercise 3 — Historical ES

Use the same return sample as in Exercise 5 and portfolio value €50 million.

1. Identify the tail observations used to compute ES$_{0.95}$ (based on your VaR rule from Exercise 2).
2. Compute the **95% historical Expected Shortfall** in euros.
3. Explain in one sentence how ES reacts if the single worst return becomes even worse (e.g. from -3.2% to -8%).